27 Supersymmetric Gauge Theories

27.3 Gauge-Invariant Action for General Gauge Superfields

Our experience in the previous section with supersymmetric Abelian gauge theories suggests immediately that in a general non-Abelian gauge theory the kinematic Lagrangian for the fields $V^A_\mu(x)$, $\lambda^A(x)$, and $D^A(x)$ should appear as part of the gauge-invariant generalization of Eq. (27.2.6):

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}\sum_A f_{A\mu\nu} f_A^{\mu\nu} - \frac{1}{2}\sum_A \left(\bar\lambda_A \not{D} \lambda_A\right) + \frac{1}{2}\sum_A D_A D_A\,. \tag{27.3.1}$$

We are now using a basis for the Lie algebra with totally antisymmetric structure constants, and we are consequently not preserving the distinction between upper and lower group indices, writing all indices A, B, etc. as subscripts. Also, $f_{A\mu\nu}$ is the gauge-covariant field-strength tensor

$$f_{A\mu\nu} = \partial_\mu V_{A\nu} - \partial_\nu V_{A\mu} + \sum_{BC} C_{ABC} V_{B\mu} V_{C\nu}\,, \tag{27.3.2}$$

and $D_\mu\lambda$ is the gauge-covariant derivative of the gaugino field, which in the adjoint representation is

$$(D_\mu\lambda)_A = \partial_\mu\lambda_A + \sum_{BC} C_{ABC} V_{B\mu}\lambda_C\,. \tag{27.3.3}$$

The question is: does Eq. (27.3.1) yield a supersymmetric action?

Since the Lagrangian density (27.3.1) is manifestly gauge-invariant, we can test whether the action is supersymmetric in any convenient gauge. To find out whether $\delta\mathcal{L}_{\text{gauge}}$ is a derivative at some point X^μ, it is convenient to adopt a specific version of Wess–Zumino gauge in which $V^\mu_A(X) = 0$. Then at X the changes in the component fields are given by Eqs. (26.2.15)–(26.2.17) at $x = X$ as

$$\delta V_{A\mu} = (\bar\alpha\gamma_\mu\lambda_A)\,, \tag{27.3.4}$$

$$\delta\lambda_A = \left(\frac{1}{4} f_{A\mu\nu}[\gamma^\nu,\gamma^\mu] + i\gamma_5 D_A\right)\alpha\,, \tag{27.3.5}$$

$$\delta D_A = i\left(\bar\alpha\gamma_5\not{\partial}\lambda_A\right). \tag{27.3.6}$$

(We must set x^μ in these expressions equal to X^μ *after* calculating the change under a supersymmetry transformation, not before.) Also, the non-linear terms in $f_A^{\mu\nu}$ are quadratic in the Vs and therefore at $x = X$ have zero variation, so at $x = X$

$$\delta f_{A\mu\nu} = (\bar\alpha(\gamma_\nu\partial_\mu - \gamma_\mu\partial_\nu)\lambda_A)\,. \tag{27.3.7}$$

With one exception, the terms in Eq. (27.3.1) and their transformations under supersymmetry transformations are thus just a number of copies (labelled by A) of the Abelian theory discussed in the previous section, and therefore give a supersymmetric action. The one exception that might disturb the supersymmetry of the action arises from the second term in the gauge-covariant derivative (27.3.3) of the gaugino field:

$$\mathcal{L}_{\lambda\lambda V} = -\frac{1}{2}\sum_{ABC} C_{ABC}\left(\bar{\lambda}_A \not{V}_B \lambda_C\right), \tag{27.3.8}$$

whose variation at $x = X$ is

$$\begin{aligned}\delta\mathcal{L}_{\lambda\lambda V} &= -\frac{1}{2}\sum_{ABC} C_{ABC}\left(\bar{\lambda}_A (\delta\not{V}_B) \lambda_C\right) \\ &= -\frac{1}{2}\sum_{ABC} C_{ABC}\left(\bar{\lambda}_A \gamma_\mu \lambda_C\right)\left(\bar{\alpha}\gamma^\mu\lambda_B\right).\end{aligned} \tag{27.3.9}$$

We can write the product of bilinears on the right-hand side as the sum of two terms

$$\left(\bar{\lambda}_A \gamma_\mu \lambda_C\right)\left(\bar{\alpha}\gamma^\mu\lambda_B\right) = X_{ABC} + Y_{ABC}\,,$$

with

$$\begin{aligned}X_{ABC} &\equiv \frac{1}{4}\sum_{\pm}\left(\bar{\lambda}_A (1\pm\gamma_5)\gamma_\mu \lambda_C\right)\left(\bar{\alpha}\gamma^\mu(1\pm\gamma_5)\lambda_B\right), \\ Y_{ABC} &\equiv \frac{1}{4}\sum_{\pm}\left(\bar{\lambda}_A (1\pm\gamma_5)\gamma_\mu \lambda_C\right)\left(\bar{\alpha}\gamma^\mu(1\mp\gamma_5)\lambda_B\right).\end{aligned}$$

By using standard Fierz identities and the anticommutativity of the spinor fields, we have

$$\begin{aligned}&\left(\bar{\lambda}_A (1\pm\gamma_5)\gamma_\mu \lambda_B\right)\left(\bar{\alpha}(1\pm\gamma_5)\gamma^\mu\lambda_C\right) \\ &\qquad = \left(\bar{\lambda}_A (1\pm\gamma_5)\gamma_\mu \lambda_C\right)\left(\bar{\alpha}(1\pm\gamma_5)\gamma^\mu\lambda_B\right), \\ &\left(\bar{\lambda}_A (1\pm\gamma_5)\gamma_\mu \lambda_B\right)\left(\bar{\alpha}(1\mp\gamma_5)\gamma^\mu\lambda_C\right) \\ &\qquad = \left(\bar{\lambda}_C (1\pm\gamma_5)\gamma_\mu \lambda_B\right)\left(\bar{\alpha}(1\mp\gamma_5)\gamma^\mu\lambda_A\right).\end{aligned}$$

(To derive the first of these relations, we note that $[(1\pm\gamma_5)\gamma_\mu]_{\alpha\gamma}[(1\pm\gamma_5)\gamma^\mu]_{\delta\beta}$ may be thought of as the $\alpha\beta$ matrix element of a matrix depending on δ and γ, and may therefore be expanded in a linear combination of $\mathbf{1}_{\alpha\beta}$, $(\gamma^\mu)_{\alpha\beta}$, $[\gamma^\mu, \gamma^\kappa]_{\alpha\beta}$, $(\gamma_5\gamma^\mu)_{\alpha\beta}$, and $(\gamma_5)_{\alpha\beta}$. Because of the factors $(1\pm\gamma_5)$, the only term in the expansion is proportional to $[(1\pm\gamma_5)\gamma^\mu]_{\alpha\beta}$. Lorentz invariance and the presence of the other $1\pm\gamma_5$ factor tell us that this expansion takes the form

$$[(1\pm\gamma_5)\gamma_\mu]_{\alpha\gamma}[(1\pm\gamma_5)\gamma^\mu]_{\delta\beta} = k[(1\pm\gamma_5)\gamma_\mu]_{\alpha\beta}[(1\pm\gamma_5)\gamma^\mu]_{\delta\gamma}.$$

To determine the proportionality constant k, we may contract both sides with $(\gamma_\nu)_{\gamma\alpha}$ and find $k = -1$. The minus sign is cancelled by a minus sign arising from the anticommutation of λ_C and $\bar{\alpha}$. The other Fierz identity is proved in the same way, except that we also need to use the symmetry property (26.A.7) for Majorana bilinears.) Hence X_{ABC} is symmetric under interchange of B and C, while Y_{ABC} is symmetric under interchange of A and B. Since C_{ABC} is totally antisymmetric, both X_{ABC} and Y_{ABC} make a vanishing contribution to the sum in Eq. (27.3.9), leaving us $\delta\mathcal{L}_{\lambda\lambda V} = 0$, so that Eq. (27.3.1) gives a supersymmetric action, as was to be shown.

We can understand *why* Eq. (27.3.1) gives a supersymmetric action by identifying the superfield that has $f_{A\mu\nu}$, λ_A, and D_A as component fields. Recall that under a generalized gauge transformation, the vector superfield $V_A(x,\theta)$ has the transformation property (27.1.12):

$$\begin{aligned}\exp\left(-2\sum_A t_A V_A(x,\theta)\right) \longrightarrow \exp\left(-i\sum_A t_A \Omega_A(x,\theta)\right)\exp\left(-2\sum_A t_A V_A(x,\theta)\right)\\ \times\exp\left(+i\sum_A t_A \Omega_A^*(x,\theta)\right),\end{aligned} \tag{27.3.10}$$

where $\Omega_A(x,\theta)$ is a general left-chiral superfield. This is not a gauge-covariant transformation rule, because $\Omega_A^* \neq \Omega_A$. To eliminate the factor involving Ω_A^*, we note that Ω_A^* is a right-chiral superfield, so that $\mathcal{D}_{L\alpha}\Omega_A^* = 0$, and therefore

$$\begin{aligned}&\exp\left(-2\sum_A t_A V_A(x,\theta)\right)\mathcal{D}_{L\alpha}\exp\left(+2\sum_A t_A V_A(x,\theta)\right)\\ &\longrightarrow \exp\left(-i\sum_A t_A \Omega_A(x,\theta)\right)\exp\left(-2\sum_A t_A V_A(x,\theta)\right)\\ &\quad\times\mathcal{D}_{L\alpha}\left[\exp\left(+2\sum_A t_A V_A(x,\theta)\right)\exp\left(+i\sum_A t_A \Omega_A(x,\theta)\right)\right].\end{aligned} \tag{27.3.11}$$

This is still not gauge-covariant, because the left-superderivative $\mathcal{D}_{L\alpha}$ acts on $\exp(+i\sum_A t_A\Omega_A(x,\theta))$ as well as on $\exp(+2\sum_A t_A V_A(x,\theta))$. This is eliminated if we follow the lead of the Abelian theory discussed in the previous section, and define a spinor superfield

$$\begin{aligned}2\sum_A t_A W_{AL\alpha}(x,\theta) \equiv \sum_{\beta\gamma}\epsilon_{\beta\gamma}\mathcal{D}_{R\beta}\mathcal{D}_{R\gamma}\Bigg[\exp\left(-2\sum_A t_A V_A(x,\theta)\right)\\ \times\mathcal{D}_{L\alpha}\exp\left(+2\sum_A t_A V_A(x,\theta)\right)\Bigg].\end{aligned} \tag{27.3.12}$$

Because the product of any three $\mathcal{D}_R$s vanishes, $W_{AL\alpha}$ is left-chiral

$$\mathcal{D}_{R\beta}W_{AL\alpha}(x,\theta) = 0\,, \tag{27.3.13}$$

and because $\mathcal{D}_{R\beta}\mathcal{D}_{R\gamma}\mathcal{D}_{L\alpha}\Omega_A \propto \mathcal{D}_{R\delta}\Omega_A = 0$, $W_{AL\alpha}$ is gauge-covariant in the sense that, for a generalized gauge transformation,

$$\begin{aligned}\sum_A t_A W_{AL\alpha}(x,\theta) \longrightarrow \exp\left(-i\sum_A t_A\Omega_A(x,\theta)\right)\sum_A t_A W_{AL\alpha}(x,\theta)\\ \times\exp\left(+i\sum_A t_A\Omega_A(x,\theta)\right).\end{aligned} \tag{27.3.14}$$

To calculate the spinor superfield at a point $x^\mu = X^\mu$, we can again adopt a version of Wess–Zumino gauge in which $V_A(X) = 0$, and after a straightforward calculation find that in this gauge

$$\begin{aligned} W_{AL}(X,\theta) = \lambda_{AL}(X_+) &+ \frac{1}{2}\gamma^\mu\gamma^\nu\theta_L\left(\partial_\mu V_{A\nu}(X_+) - \partial_\nu V_{A\mu}(X_+)\right) \\ &+ \left(\theta_L^T\epsilon\theta_L\right)\not{\partial}\lambda_{RA}(X_+) - i\theta_L D_A(X_+). \end{aligned}$$

Since W_{AL} is gauge-covariant, at a general point in a general gauge it must have the value

$$W_{AL}(x,\theta) = \lambda_{AL}(x_+) + \frac{1}{2}\gamma^\mu\gamma^\nu\theta_L f_{A\mu\nu}(x_+) + \left(\theta_L^T\epsilon\theta_L\right)\not{D}\lambda_{RA}(x_+) - i\theta_L D_A(x_+). \tag{27.3.15}$$

From this, we can construct a Lorentz- and gauge-invariant $\mathcal{F}$-term bilinear in W

$$\begin{aligned} -\left[\sum_{A\alpha\beta}\epsilon_{\alpha\beta}W_{AL\alpha}W_{AL\beta}\right]_{\mathcal{F}} = \sum_A\Big[&-\left(\bar{\lambda}_A\not{D}(1-\gamma_5)\lambda_A\right) - \frac{1}{2}f_{A\mu\nu}f_A^{\mu\nu} \\ &+ \frac{i}{4}\epsilon_{\mu\nu\rho\sigma}f_A^{\mu\nu}f_A^{\rho\sigma} + D_A^2\Big]. \end{aligned} \tag{27.3.16}$$

Just as in the previous section, the gauge-invariant Lagrangian (27.3.1) is obtained from the real part of this $\mathcal{F}$-term:

$$-\frac{1}{2}\,\mathrm{Re}\left[\sum_{A\alpha\beta}\epsilon_{\alpha\beta}W_{AL\alpha}W_{AL\beta}\right]_{\mathcal{F}} = \mathcal{L}_{\text{gauge}}\,. \tag{27.3.17}$$

What about the imaginary part? This is given by

$$\begin{aligned} -\,\mathrm{Im}\left[\sum_{A\alpha\beta}\epsilon_{\alpha\beta}W_{AL\alpha}W_{AL\beta}\right]_{\mathcal{F}} = &-i\sum_A\left(\bar{\lambda}_A\not{D}\gamma_5\lambda_A\right) \\ &+ \frac{1}{4}\epsilon_{\mu\nu\rho\sigma}\sum_A f_A^{\mu\nu}f_A^{\rho\sigma}. \end{aligned} \tag{27.3.18}$$

Eq. (26.A.7) and the antisymmetry of the structure constants show that $(\bar{\lambda}_A\not{D}\gamma_5\lambda_A) = \frac{1}{2}\partial_\mu(\bar{\lambda}_A\gamma^\mu\gamma_5\lambda_A)$, so the first term is a total derivative, while Eq. (23.5.4) tells us that the second term is a total derivative also. In Abelian gauge theories, this means that a term like (27.3.18) would have no effect. But as discussed in Sections 23.5 and 23.6, in non-Abelian gauge theories the existence of instanton solutions allows the density (27.3.18) to have a non-vanishing integral over spacetime. We therefore must allow for the possibility of a new term in the Lagrangian density

$$\mathcal{L}_\theta = -\frac{g^2\theta}{16\pi^2}\,\mathrm{Im}\left[\sum_{A\alpha\beta}\epsilon_{\alpha\beta}W_{AL\alpha}W_{AL\beta}\right]_{\mathcal{F}}\,, \tag{27.3.19}$$

where θ is a new real parameter, and g is a gauge coupling, which can be conveniently defined for a simple gauge group so that if t_A, t_B, and t_C are in the 'standard' $SU(2)$ subalgebra of the gauge algebra used in calculating instanton effects, we have $C_{ABC} = g\epsilon_{ABC}$. With this definition of the gauge coupling, Eq. (23.5.20) gives for simple gauge groups

$$\int d^4x\, \epsilon_{\mu\nu\rho\sigma} \sum_A f_A^{\mu\nu} f_A^{\rho\sigma} = 64\pi^2\nu/g^2\,, \tag{27.3.20}$$

where $\nu = 0, \pm 1, \pm 2, \ldots$ is an integer, the winding number, which characterizes the topological class of the gauge field configuration. Thus for instantons of winding number ν, the Lagrangian density $\mathcal{L}_\theta$ contributes a phase to path integrals, given by

$$\left[\exp\left(i\int d^4x\, \mathcal{L}_\theta\right)\right]_\nu = \exp(i\nu\theta)\,, \tag{27.3.21}$$

so the effects of $\mathcal{L}_\theta$ are periodic in θ, with period 2π.

It is often convenient to absorb a factor g into the gauge field, so that the structure constants do not depend on g, and instead the Lagrangian density for the gauge field is multiplied by an over-all factor $1/g^2$. In this notation, the complete Lagrangian density for the gauge field may be written in terms of the rescaled gauge fields and structure constants as

$$\mathcal{L}_{\rm gauge} + \mathcal{L}_\theta = -\,{\rm Re}\left[\frac{\tau}{8\pi i}\sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{AL\alpha} W_{AL\beta}\right]_{\mathcal{F}}\,, \tag{27.3.22}$$

where τ is the complex coupling parameter

$$\tau = \frac{4\pi i}{g^2} + \frac{\theta}{2\pi}\,. \tag{27.3.23}$$

According to Eq. (23.5.19), the contribution of instantons of winding number ν to path integrals is suppressed by a factor $\exp(-8\pi^2|\nu|/g^2)$, which together with the factor (27.3.21) yields an over-all factor

$$\exp\left[i\nu\theta - \frac{8\pi^2|\nu|}{g^2}\right] = \begin{cases} \exp(2\pi i\nu\tau), & \nu \geq 0, \\ \exp(2\pi i\nu\tau^*), & \nu \leq 0. \end{cases} \tag{27.3.24}$$